

Introduction

1. COE 608
2. OPERATING SYSTEM

COE718: Embedded Systems Design

<http://www.ecb.torontomu.ca/~courses/coe718/>

Dr. Gul N. Khan

<http://www.ecb.torontomu.ca/~gnkhan>

Electrical, Computer and Biomedical Engineering

Toronto Metropolitan University

Overview

Section 1 → 6
7, 8 & 10

- Course Management
- Embedded Systems and Applications
- Characteristics of Embedded Systems
- Course Contents

Electrical, Computer and Biomedical Engineering

COE718: Embedded Systems Design

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Consultation: Wednesday 11:45AM-1:00PM & after Lecture

2012

**Department of Electrical, Computer & Biomedical Engineering
Toronto Metropolitan University**

Lectures and Projects

Half Notes

SLIDES

- You will need to take notes from lectures and also require text-reference books and some research articles identified by the instructor.

Labs and Project

- Aimed at concept reinforcement and practical experience.

Lectures, Labs, Projects and other support material will be available at the course website:

⇒ <http://www.ecb.torontomu.ca/~courses/coe718/>

OPENLY AVAILABLE
ANNOTATED SLIDES
ON D2L

Assessment and Evaluation

Labs: 20% (4+4+4+4+4)

Project: 15%

Midterm Exam: 25% (Wednesday, October 15)

Final Exam: 40%

ENG 411 (PHYSICAL HARDWARE)

8:15AM

SOMETIME IN DEC

Course Text-Reference Books and other Material



Daniel W. Lewis, Fundamentals of Embedded Software with the ARM Cortex M3, Pearson 2013 ISBN 978-0-13-291654-7 60 → 70 %

Other Reference Books

- *T. Martin*, The Designer's Guide to the Cortex-M Processor Family: A Tutorial Approach, Elsevier, 2013, ISBN 978-0080982960
- ? • *Marilyn Wolf*, Computer as Components: Principles of Embedded Computing System Design, 4th Edition Morgan Kaufman - Elsevier 2016, ISBN 978-0-12-805387-4
- Embedded Core Design with FPGAs by *Z. Navabi*, McGraw Hill 2007, ISBN 978-0-07-147481-8 or ISBN 0-07-147481-1
- *• *Alan Burns and Andy Wellings*, Real-time Systems & Programming Languages, Addison-Wesley 2001, ISBN 0 201 72988 1 ✓

Some Articles, Embedded Processors and other Data Sheets are available at the Course Website:

<http://www.ecb.torontomu.ca/~courses/coe718/>

Lecture's Material

- **Introduction to Embedded Systems**

Text by D. Lewis: Chapter-1, Professional Journal and Magazine Articles

- **Embedded CPUs and Cores**

ARM Cortex M3 Core Architecture: Chapter 5 of Text by D. Lewis.

- **ARM Cortex M3 ISA and Programming**

Text by D. Lewis: part of Chapters 6 and 7.

- **Real-time Operating Systems**

Concurrency, pre-emptive and non-pre-emptive Scheduling. Keil RTX operating system environment: Chapter 9 of Text by D. Lewis

- **Real-time Scheduling**

FPS and EDF Real-time Scheduling techniques, RTX real-time operating environment. Chapter 10 of Text by D. Lewis and Part of Chap. 7-10, 13 of Text by Burns & Wellings

- **Introduction to Hardware Software Codesign**

Embedded System Modeling and Co-specification

Text by M. Wolf, part of Chapter 7; Journal Articles, Manuals

- **Embedded System Partitioning, Accelerator based Embedded Systems**

Text by M. Wolf: part of Chapters 7 and 8; Review Journal Articles

- **Fault-tolerant Embedded Systems - Review Articles**

- **Embedded System Case Studies – If time permits**

COE838
SYSTEM ON CHIP DESIGN

⇒ A9, → A75

ARM M3/M4

COE 538
MICROPROCESSOR SYSTEM

68HC12
68HC11

Introduction

- What are Embedded Systems?
- Challenges in Embedded Computing System Design
- Design Methodologies

TRAFFIC
LIGHT
controller

EMBEDDED COMPUTER
SYSTEM

SOFTWARE & HARDWARE

Computer Org.
& ARCH.

RTOS
Real-time operating
SYSTEM

Main Aim of the Course

- To introduce the basics of embedded system design.
 - Software and hardware components of an embedded system
- Hardware-Software Codesign
- To understand real-time operating systems
- Embedded Computer Architecture

Computer
RTX from ARM
C/C++

Ideally Student should have the knowledge of:

ASSEMBLER
for ARM M3/M4

COE 608

- Basics of Programming C or C++ and Computer Architectures
- Operating Systems

COE 628

What is an Embedded (Real-time) System?

Most embedded systems are also real-time systems.

- An embedded system is an information processing system that responds to externally generated input stimuli within a finite and specified period.
 - ✓ ■ The correctness depends not only on the logical result but also the time it was delivered
 - Failure to respond is as bad as the wrong response!
- Embedded system: any device that includes a programmable computer but is not itself a general-purpose computer.
- Take advantage of application characteristics to optimize the design:

Don't need all the general-purpose bells and whistles.

AEROSPACE / MILITARY APP.

MSEL → 1-2 msec

MEET DEADLINE

HARD MISSILE CONT.
SOFT VIDEO PLAYER

RESPONSE TIME

40 msec
one frame

MP4 VIDEO PLAYER

25/30 FRAMES
per second

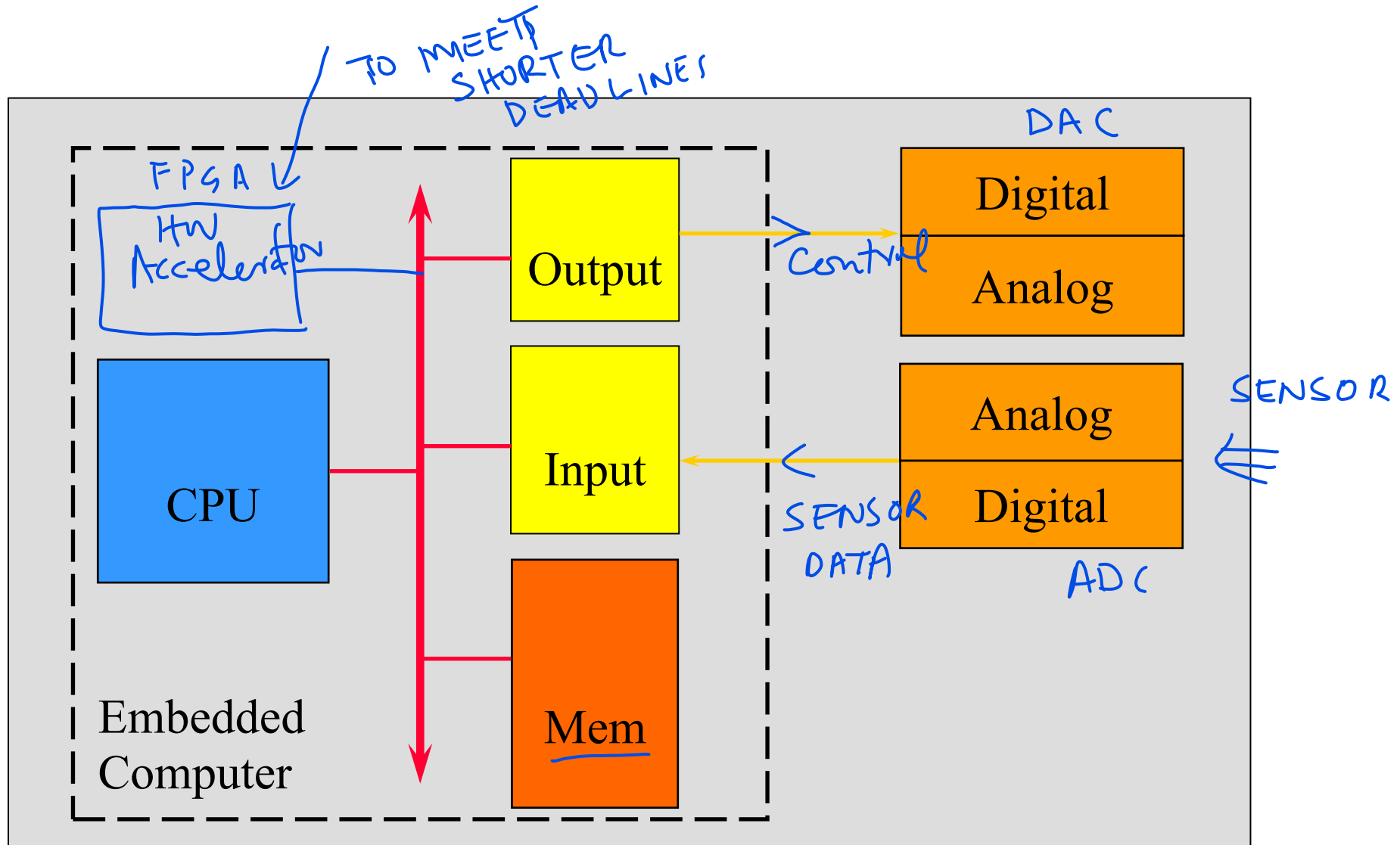
one frame 150 msec

(CPI + HARDWARE)

HARDWARE-SOFTWARE
CODESIST

Accelerator HARDWARE

Embedding a Computer



Embedded Systems

- Electronic devices that incorporate a computer (microprocessor/micro-controller) in their implementation.
- A computer is used in such devices primarily as a means to simplify the system design and to provide flexibility.
- Often the user of the device is not even aware that a computer is present.
- Embedded Systems are every where
 - Embedded processors account for more than 90% of worldwide microprocessor production.
 - Embedded vs. desktop = 100:1
 - 99% of all processors are for the embedded systems market.
 - Number of embedded processors in a typical home is estimated at 50-60.
(a typical automobile has more than 50 embedded processors)

Embedded Real Time Systems

- Real-time systems process events.
- Events occurring on external inputs cause other events to occur as outputs.
- Minimizing response time is usually a primary objective, or otherwise the entire system may fail to operate properly.

Types of Embedded Real Time System

- ① • Hard real-time — e.g. Flight control systems.
- ② • Soft real-time — e.g. Data acquisition system.
- ①a • Real real-time — e.g. Missile guidance system. PART OF ①
MUCH SMALLER RESPONSE TIME
- ②a • Firm real-time

Types of Real Time System

- **Hard real-time** — systems where it is absolutely imperative that the responses occur within the required deadline.

For example: Flight control systems.

- **Soft real-time** — systems where deadlines are important, but which will still function correctly if deadlines are occasionally missed. For example: Data acquisition system.

- **Real real-time** — systems which are hard real-time, and their response time is very short.

For example: Missile guidance system.

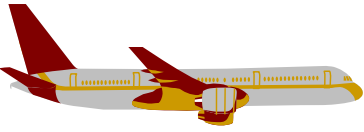
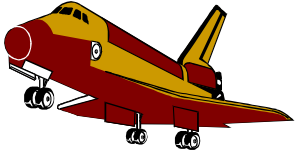
- **Firm real-time** — systems which are soft real-time but in which there is no benefit from late delivery of service.

A single embedded system may have hard, soft and real real-time subsystems.

In reality many systems will have a cost function associated with the missing of each deadline.

SPACE STATION } 20 million
control } LINE OF
CODE

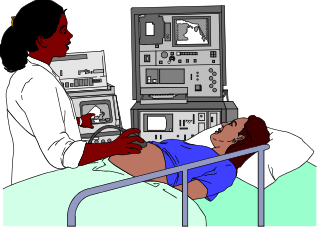
Embedded System Applications

Aerospace Spacecrafts 	Navigation systems, automatic landing systems, flight attitude controls, engine controls, space exploration (e.g., the <u>Mars</u> Pathfinder). 
<u>Automotive</u>	Fuel injection control, <u>passenger environmental controls</u>, <u>anti-lock braking</u>, <u>air bag controls</u>, <u>GPS mapping</u>.
Children's Toys	Nintendo's, Mattel's "My Interactive Pooh", Tiger Electronics' "Furby".
Communi- cations	Satellites; network routers, switches, hubs. <i>Modems</i>

Embedded System Applications

Computer Peripherals	Printers, scanners, keyboards, displays, modems, hard disk drives, CD-ROM drives.
Home	Dishwashers, microwave ovens, HDTV, sound systems, fire/security alarm systems, lawn sprinkler controls, thermostats, cameras, clock digital radios. PATIENT MONITORING
Industrial	Elevator controls, surveillance systems, robots. ↑ Reliability / FAULT-TOLERANT COMPUTERS
Instrumentation	Data collection, oscilloscopes, signal generators, signal analyzers, power supplies.

Embedded System Applications

Medical  An illustration of a doctor in a white coat standing next to a patient lying in a hospital bed. The doctor is holding a device, possibly a stethoscope or a small monitor, and looking at the patient. In the background, there is a large medical monitor displaying various waveforms and data.	Imaging systems (e.g., XRAY, MRI, and ultrasound), patient monitors, and <u>heart pacers</u>.
Office Automation	Copiers/ FAX machines, smart telephones, and cash registers.
Personal	Tablets, ipads, cell phones, smart-watches

Early History of Embedded Systems

- First microprocessor was Intel 4004 in early 1970's.
- HP-35 calculator used several chips to implement a microprocessor in 1972.
- Automobiles used microprocessor-based engine controllers starting in 1970's.
 - Control fuel/air mixture, engine timing, etc.
 - Multiple modes of operation: warm-up, cruise, etc.
 - Provides lower emissions, better fuel efficiency.
- Microcontroller: includes I/O devices, on-chip memory.
- Digital signal processor (DSP): microprocessor optimized for digital signal processing.

Typical embedded word sizes: 8-bit, 16-bit, and 32-bit.

64-bit

military
TRUCK
ENGINE
controls

Automotive embedded systems

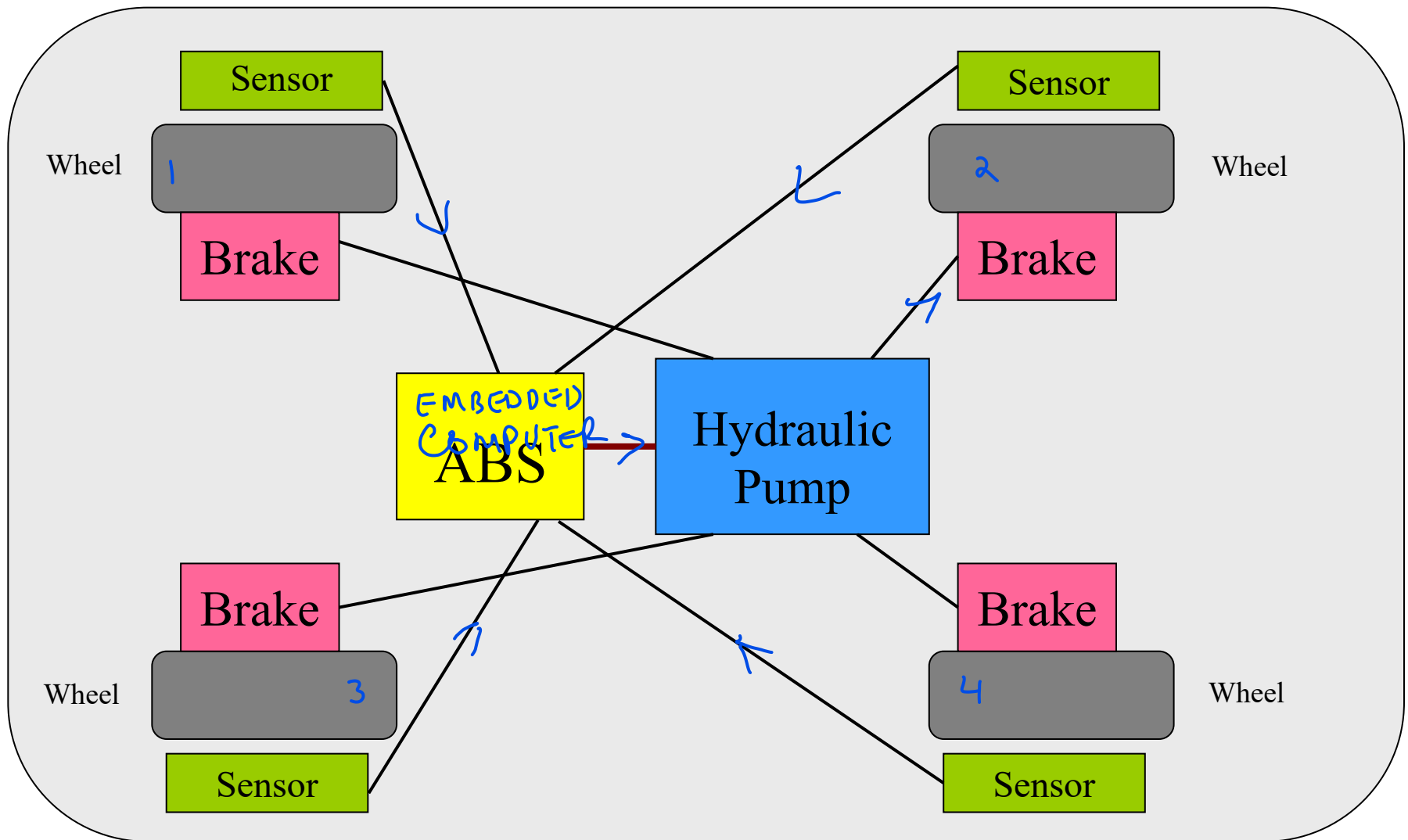
Today's high-end automobile may have around 100 microprocessors:

- 4-bit microcontroller checks seat belt
- Microcontrollers run dashboard devices
- 16/32-bit microprocessor controls engine

BMW brake and stability control system

- Anti-lock brake system (ABS): Pumps brakes to reduce skidding.
- Automatic Stability Control (ASC+T): Controls engine to improve stability.
- ABS and ASC+T communicate.
ABS was introduced first---needed to interface to existing ABS module.

Anti-lock Brake System (ABS)



Embedded System Applications

Programmable Digital Thermostat

Uses: 4-bit Microprocessor



Vending Machine



Vendo V-MAX 720

Uses
8-bit Motorola 68HC11
Microcontroller

Vitality's GlowCap



Vitality's GlowCap

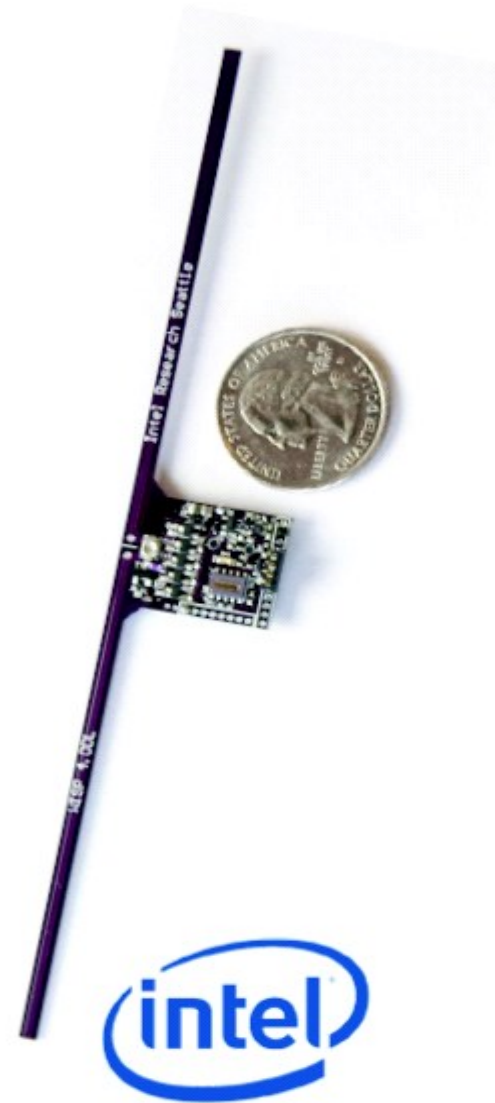
- GlowCap has a tiny Amtel 8-bit picoPower AVR Processor
- Help People to take their medication on-time.
- Sense when the bottle is opened.
- Connect to Vitality server and transmit information wirelessly.



Intel WISP RFID

TIMSP430F1232: Low Power Micro-controller

- 16-bit CPU
- 8 KBytes of flash memory
- 256 bytes of RAM
- 10-bit –ADC with 200 kilo-samples/second
- CPU can run at 8MHz with 3.3V supply voltage



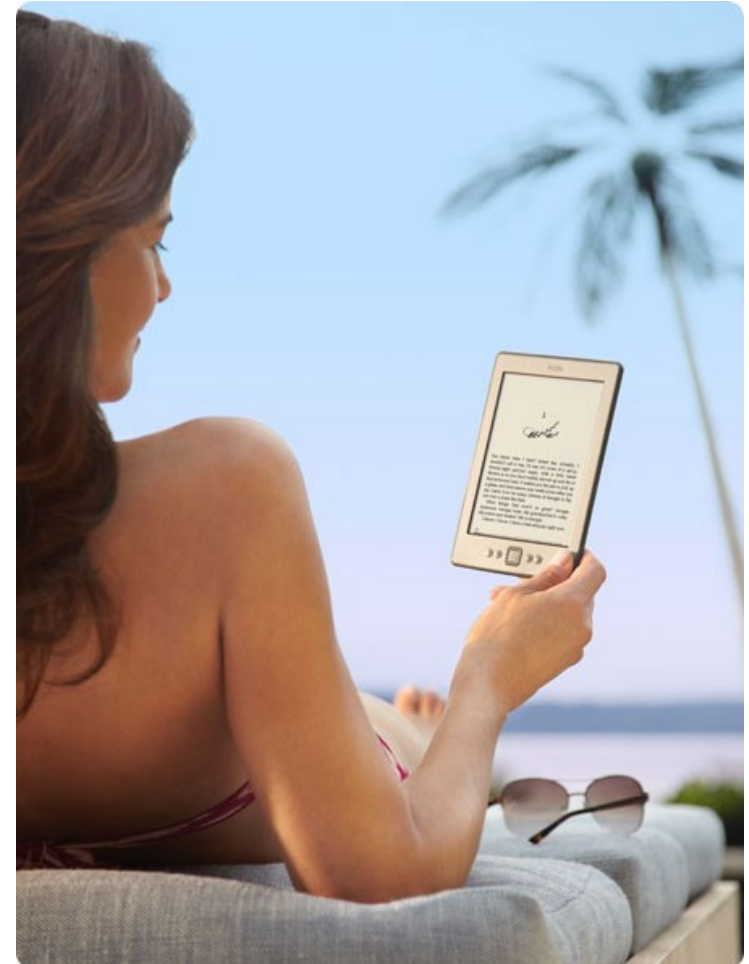
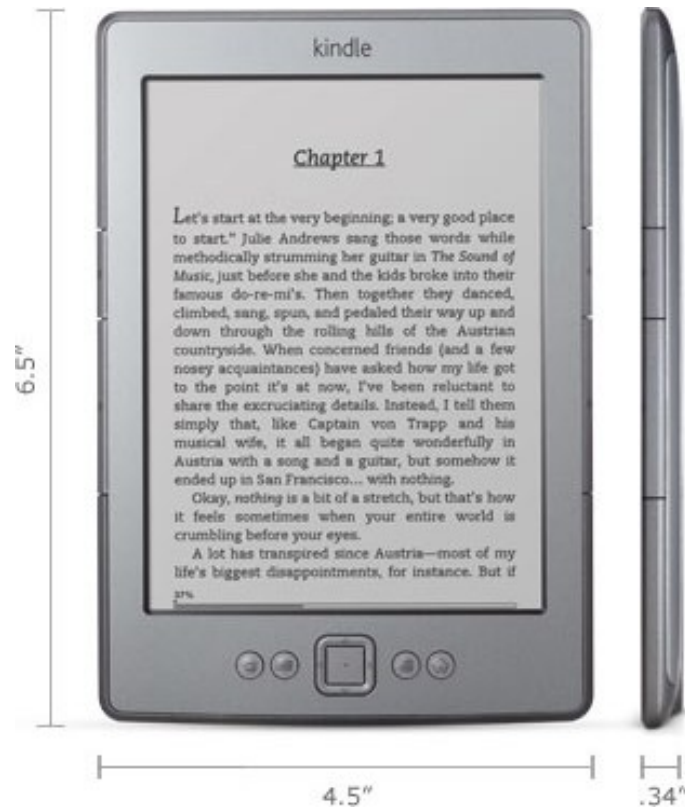
Sonicare Diamond Clean Toothbrush

Toothbrush
8-bit PIC
microprocessor



Amazon Kindle-2

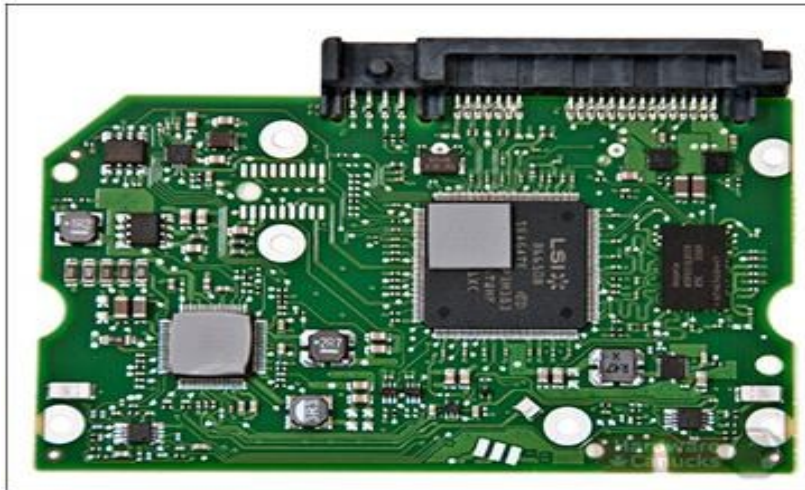
Uses 32-bit
ARM
processor



Reads like real paper — no glare, even in bright sunlight

Seagate Barracuda Disk Drive

- Employ two ARM Cortex-R4 Processors
- One for Servo control and 2nd for Command and Data Flow



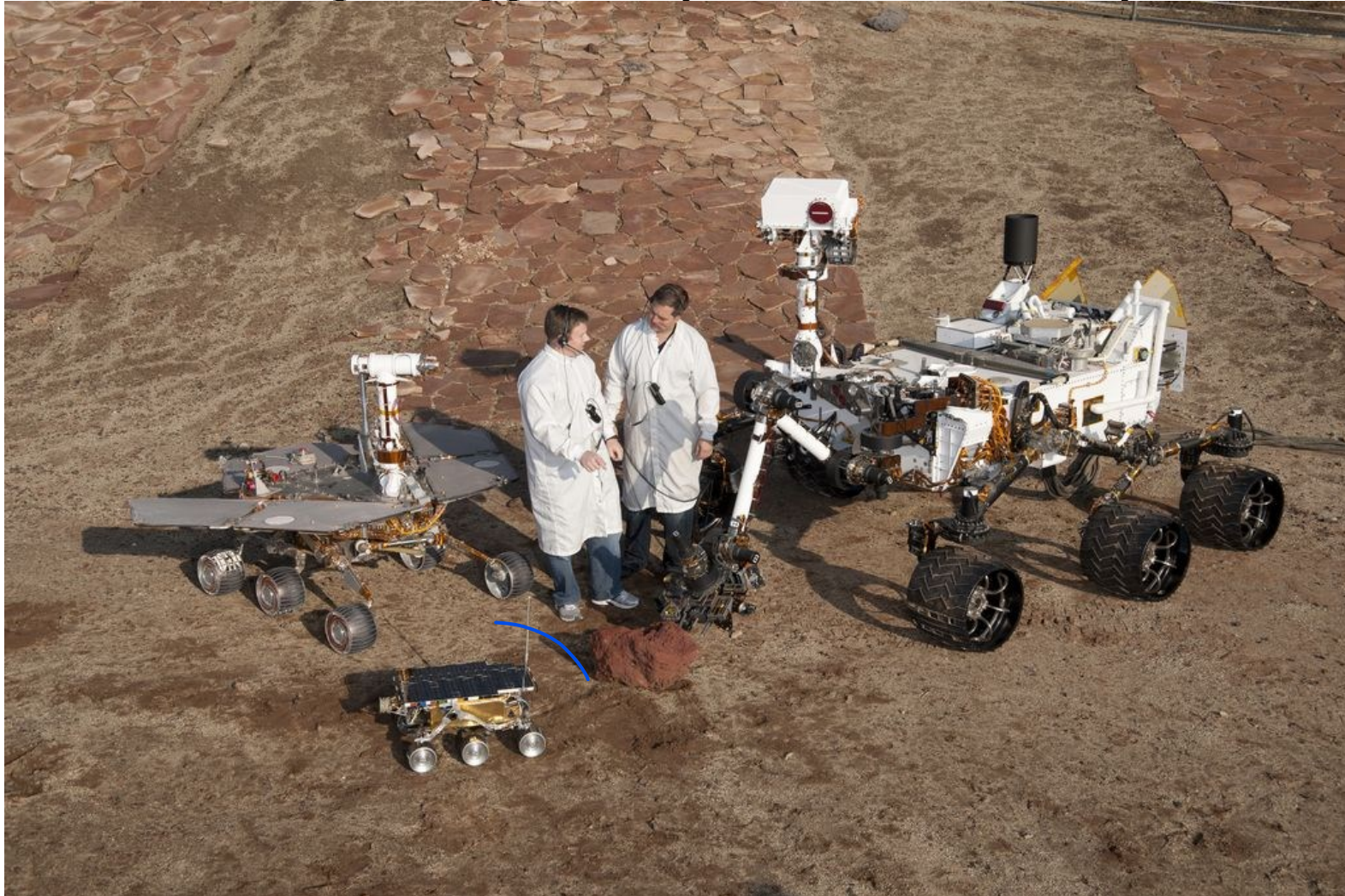
NASA's Mars Rover

PathFinder: Uses Intel 80C85 8-bit Microprocessor



MAR's Rovers

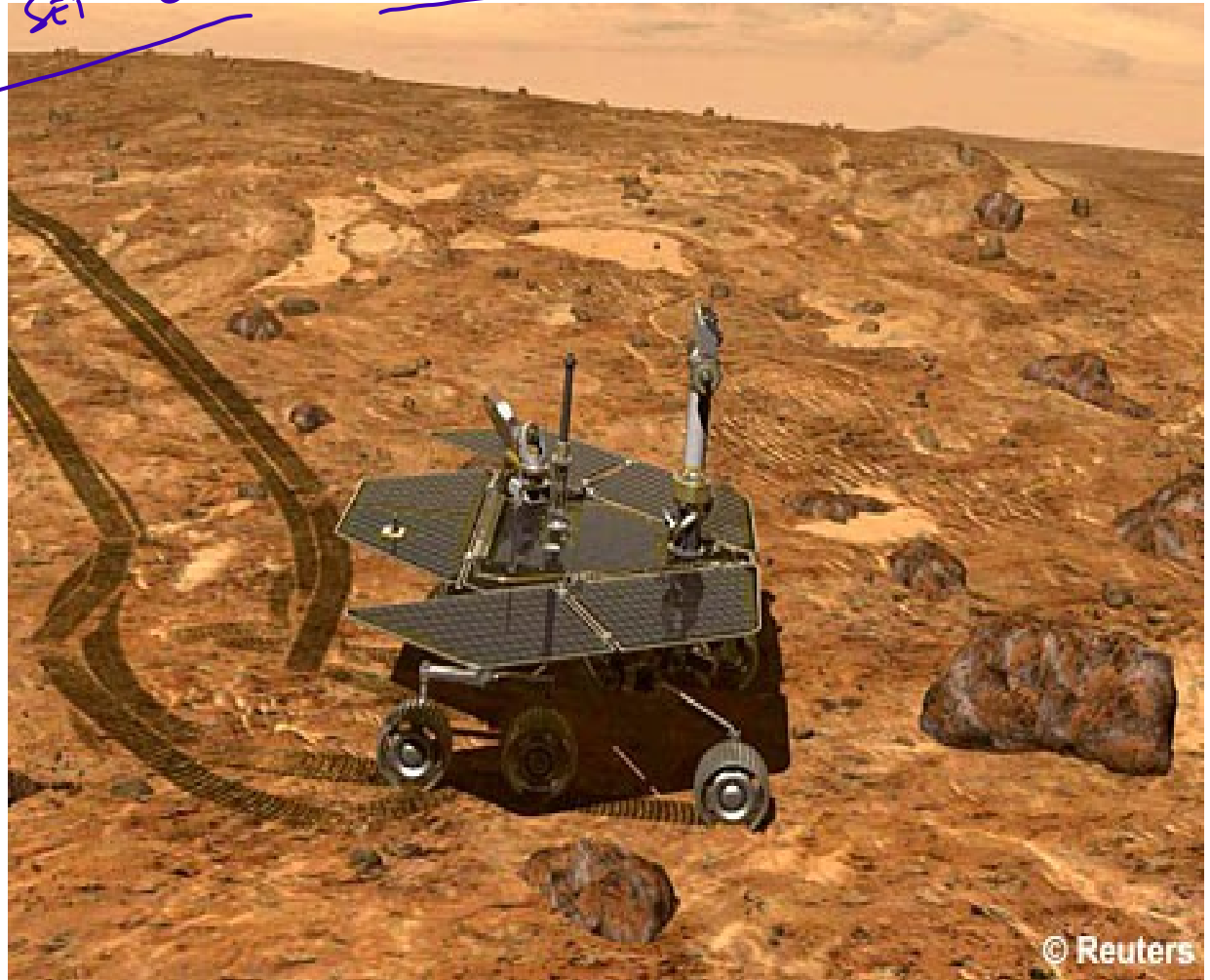
Pathfinder-1997, Spirit/Opportunity-2003 and Curiosity-2012



2003 MAR'S Rover

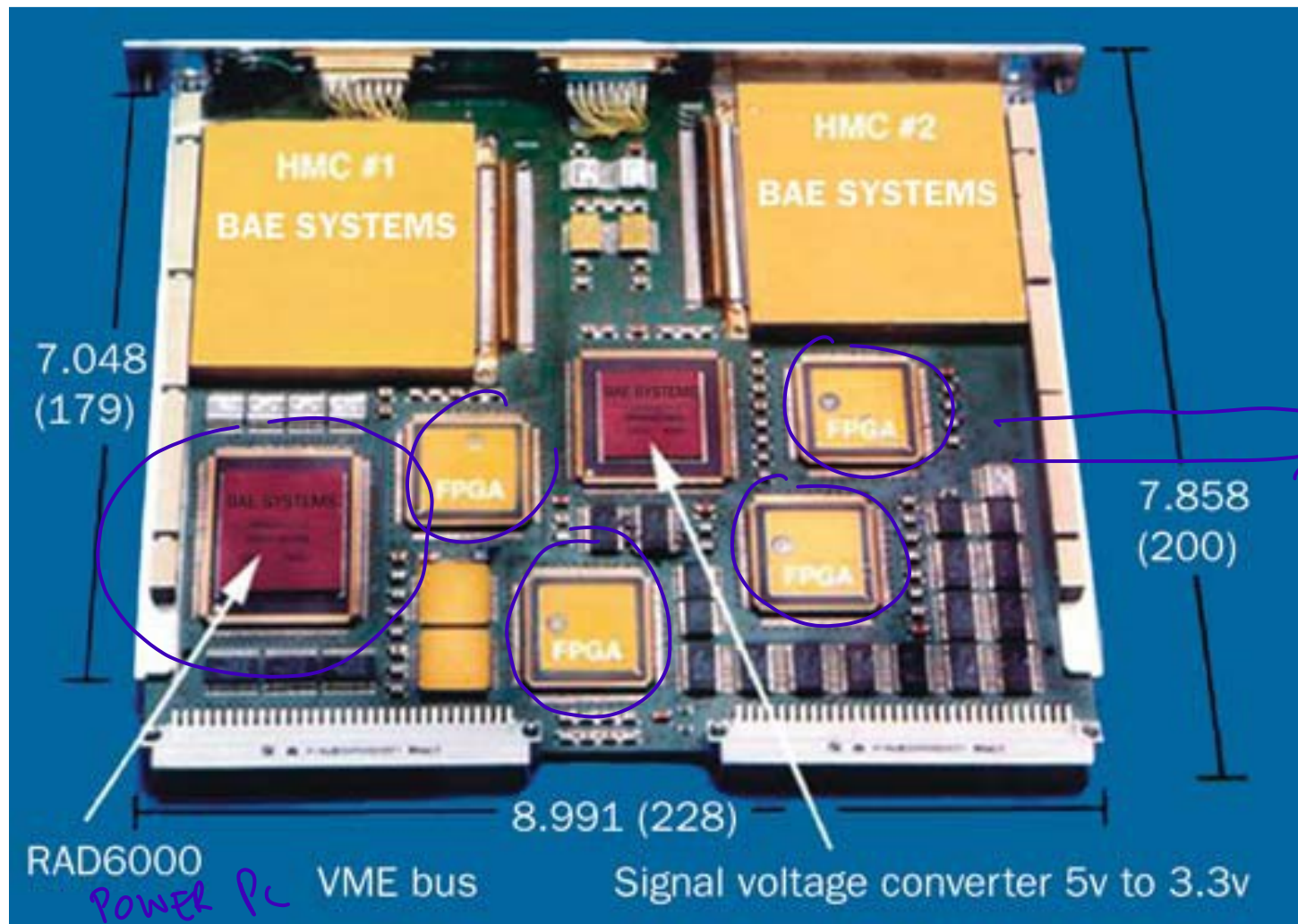
Spirit/Opportunity

- Use BAE Systems RAD6000 32-bit RISC Processor
- Radiation hardened IBM POWER series 6000 CPU
- Employ VxWorks: Embedded Real-time Operating System from Wind River.



1997
↓
RTX

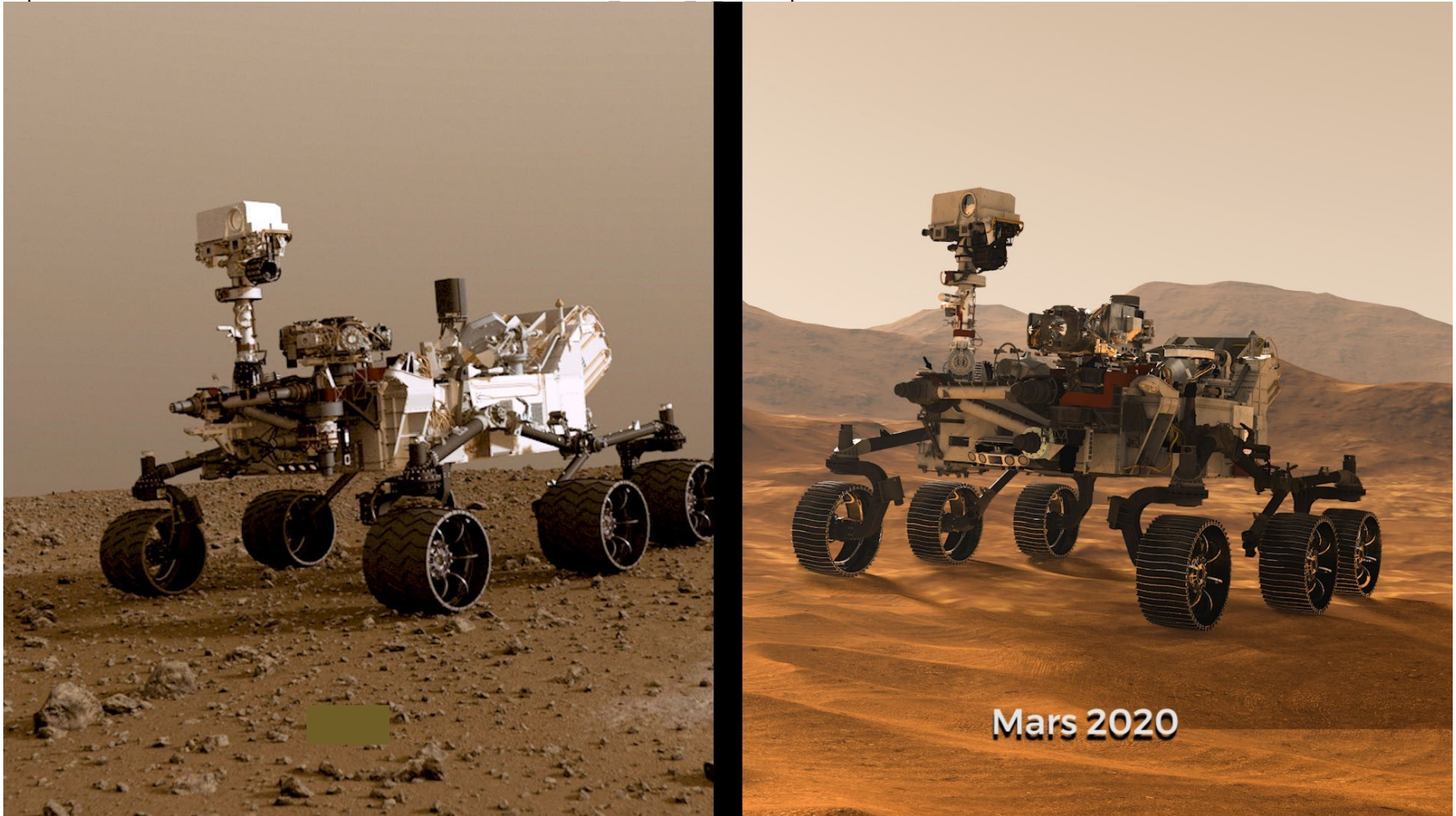
Mars Rover RAD6000 Flight Computer, FPGA-based



MARS Rover 2020 - Perseverance Rover

Landed February 2021

[http://www.ecb.torontomu.ca/~courses/coe718/lectures/Perseverance Arrives at Mars.mp4](http://www.ecb.torontomu.ca/~courses/coe718/lectures/Perseverance%20Arrives%20at%20Mars.mp4)



Comparison of Embedded Computer Systems for Mars Rovers

<u>Rover (mission, year)</u>	<u>CPU</u>	<u>RAM</u>	<u>Storage</u>	<u>Operating system</u>
<i>Sojourner</i> Rover (Pathfinder 1997)	2MHz Intel 80C85	512KB	176 KB	Custom cyclic executive
Pathfinder Lander (1997) (Base station for <i>Sojourner</i> rover)	20MHz IBM RAD6000	128 MB	6 MB (EEPROM)	VxWorks (multitasking)
<i>Spirit</i> and <i>Opportunity</i> (Mars Exploration Rover, 2004)	20 MHz IBM RAD6000	128 MB	256 MB	VxWorks (multitasking)
<i>Curiosity</i> (Mars Science Laboratory, 2011)	200 MHz IBM RAD750	256 MB	2GB	VxWorks (multitasking)
<i>Perseverance</i> 2 Compute Elements (Mars Rover, 2020) Landed 2021	200 MHz IBM RAD750 PowerPC 750	256 MB	2GB Flash Memory 256KB EEPROM	VxWorks (multitasking)

Non-functional Requirements

FUNCTIONAL

- Many embedded systems are mass-market items that must have low manufacturing costs.
- Limited memory, microprocessor power, etc.
- Power consumption is critical in battery-powered devices.
- Excessive power consumption increases system cost even in wall-powered devices.

SIZE
WEIGHT
POWER CONSUMPTION

i, j, k

1. CPU
2. MEMORY

LESS HARDWARE

Power

- Custom logic is a clear winner for low power devices.
- Modern microprocessors offer features to help control power consumption.
- Software design techniques can help reduce power consumption.

INTELLIGENT

MEMORY
EFFICIENT

READ/WRITE
MEMORY

68HC12

Acc [m]

Where are we heading?

- **Embedded CPUs and Cores**
ARM Cortex M3 Core ISA and Architecture, Chapter 5
- **ARM Cortex M3 ISA and Programming**
Part of Chapters 6 and 7.
- **Real-time Operating Systems**
Concurrency, pre-emptive and non-pre-emptive Scheduling.
Keil RTX operating system environment: Chapter 9
- **Real-time Scheduling**
FPS and EDF Real-time Scheduling techniques, RTX real-time operating environment manual and details. Chapter 10
- **Introduction to Hardware Software Codesign**
Embedded System Modeling and Co-specification
Text by M. Wolf, part of Chapter 7; Journal Articles, etc.
- **System Partitioning, Accelerator based Embedded Systems**
Text by M. Wolf: part of Chapters 7 and 8; Review Journal Articles
- **Fault-tolerant Embedded Systems - Review Articles**
- **Case Studies – if time permits**